# Advanced optimal control: from reusable rocket landing to efficient training of neural networks

IPSA course 2026

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# Course schedule

1. Optimal Control Problems (OCP): review of ordinary differential equations; existence of solutions to OCP.

2. Optimality Conditions for OCP: the Maximum Principle and structure of optimal controls; application to reusable rocket landing.

3. Python Session 1: real-world reusable rocket landing.

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→ Group evaluation: please form groups and report them in the online sheet (see website)!

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# Recap

Cost (functional)

$$\min_{u(\cdot)\in L^\infty} \; h(x(t_f)) + \int_0^{t_f} f^0(t, x(t), u(t))\, dt$$

(Control) constraints

$$u(t) \in U \subseteq \mathbb{R}^m, \quad t \in [0, t_f]$$

Dynamics + initial/final conditions

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3. <u>Today's topic</u>: characterize optimal solutions $u^*(\cdot) \in L^\infty([0, t_f], U)$ to OCP via <u>necessary conditions of optimality</u>.

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Lecture 2 - Characterize the structure of solutions to OCP

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# Today’s detailed schedule

1. Necessary conditions for optimality: the Pontryagin Maximum Principle (PMP).

2. Learn how to apply the PMP through simple optimal control problems.

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<u>Key insight</u>: OCP is a (infinite-dimensional) constrained optimization problem.

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Let us introduce such conditions for optimality for OCP, and characterize optimal controls!

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b. <u>Backward linear ODE</u>: unique solution exists

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3. <u>Theorem PMP</u> (necessary conditions for optimality).

   Let $(u^*, x^*)$ be an optimal control-trajectory pair solution to OCP. There exists a pair $(p^0, \mathfrak{p}) \in \{-1, 0\} \times \mathbb{R}^\ell$ such that the following conditions hold true:

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   b. Non-triviality: $(p^*(t), p^0) \neq 0 \text{ for all } t \in [0, t_f]$

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   Evaluated at the optimal pair!

   b. Non-triviality: $(p^*(t), p^0) \neq 0 \text{ for all } t \in [0, t_f]$

   c. Maximality condition: $u^*(t) \in \arg\max_{u \in U} H(t, x^*(t), u, p^*(t), p^0)$

# The PMP - Theory

3. Theorem PMP (necessary conditions for optimality).

   Let $(u^*, x^*)$ be an optimal control-trajectory pair solution to OCP. There exists a pair $(p^0, \mathfrak{p}) \in \{-1, 0\} \times \mathbb{R}^\ell$ such that the following conditions hold true:

   a. Adjoint and adjoint system:

   Called the adjoint

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Used to find simpler forms of $u^*(\cdot)$.

# Today’s detailed schedule

1. Necessary conditions for optimality: the Pontryagin Maximum Principle (PMP).
2. Learn how to apply the PMP through simple optimal control problems.
3. Characterize optimal controls for the reusable rocket landing problem.

# The PMP - Examples

Minimal energy double integrator - Used in electronic circuit eco-phasing

$$\min_{u(\cdot)\in L^\infty([0,t_f],[-1,1])} \int_0^{t_f} u(t)^2\, dt, \quad t_f > 0 \text{ fixed}$$

$$\begin{cases} \dot{y}(t) = z(t) \\ \dot{z}(t) = u(t) \end{cases} \quad \text{with} \quad \begin{cases} y(0) = 1, \;\; z(0) = 0 \\ y(t_f) = z(t_f) = 0 \end{cases}$$

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$$H(t, x, u, p, p^0) = p^\top f(t, x, u) + p^0 f^0(t, x, u) = p_y z + p_z u + p^0 u^2$$

# The PMP - Examples

2. Given the Hamiltonian $H(t, x, u, p, p^0) = p_y z + p_z u + p^0 u^2$, <u>there exist a vector</u> $\mathfrak{p} \in \mathbb{R}^2$ <u>and an adjoint</u> $p^* : [0, t_f] \to \mathbb{R}^2$ satisfying the adjoint system:

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In particular, we compute

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Thus:
$$p_y^*(t) = \mathfrak{p}_y, \quad p_z^*(t) = \mathfrak{p}_z + \mathfrak{p}_y(t_f - t), \quad t \in [0, t_f]$$

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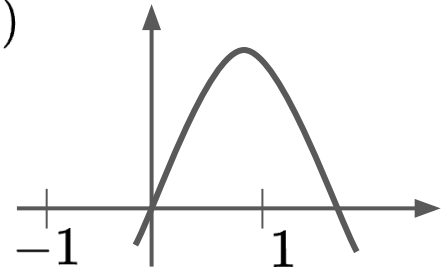

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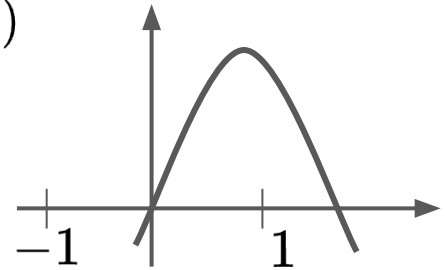
Solutions depend on the value of $p^0$ !



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Maximizing a parabola:

$$u^*(t) = \arg\max_{u \in [-1,1]} (p_z^*(t)u - u^2)$$

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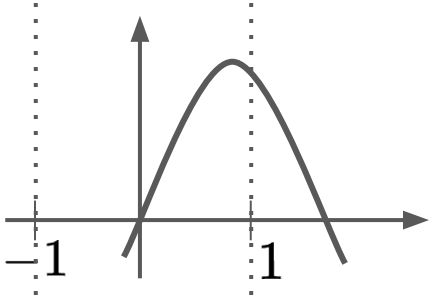

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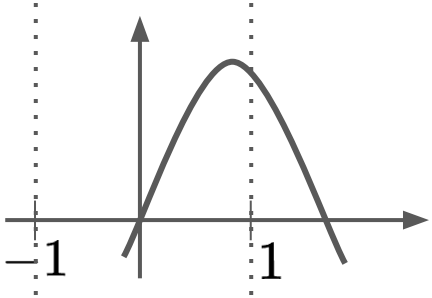

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Maximizing a straight line in [-1,1]

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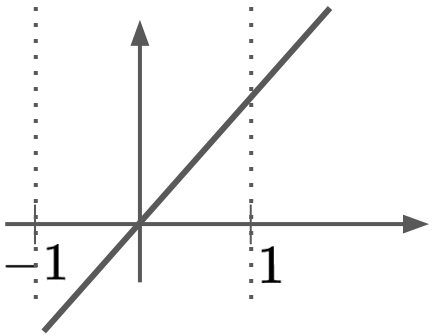

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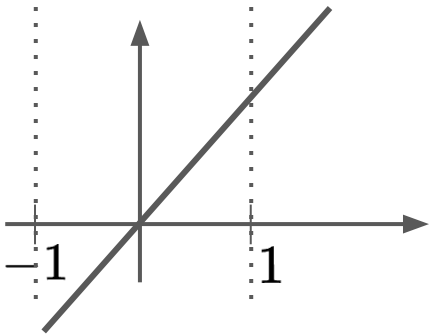

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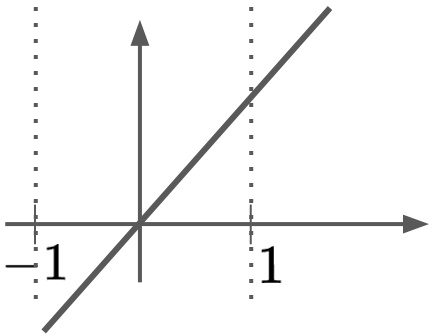
What if $p_z^*(t) = 0$ ???



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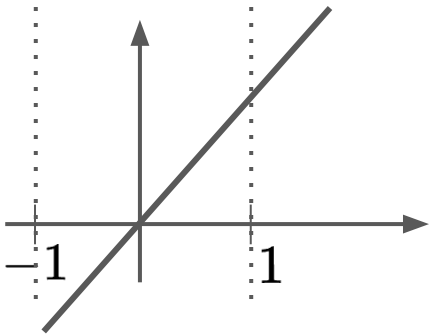


What if $p_z^*(t) = 0$ ???

Luckily, $p_z^*(t) = 0$ and $p^0 = 0$ <u>never happens</u>.

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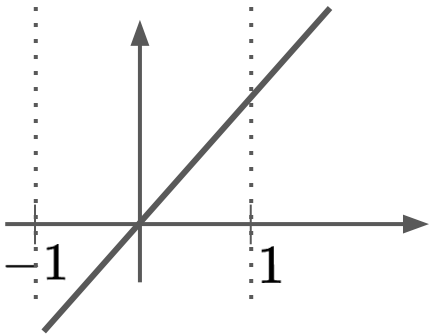


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# The PMP - Examples

Let us solve: $\arg\max_{u \in U} H(t, x^*(t), u, p^*(t), p^0) = \arg\max_{u \in [-1,1]} (p_z^*(t)u + p^0 u^2)$

b. Case $p^0 = 0$

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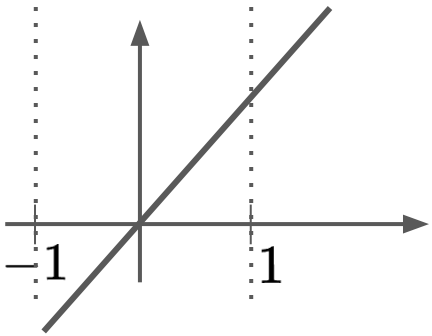


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Finally, we would obtain $p_y^*(t) = p_z^*(t) = p^0 = 0,\ t \in [0, t_f]$, contradiction!

# The PMP - Examples

Minimal energy double integrator - Used in electronic circuit eco-phasing

$$\min_{u(\cdot)\in L^\infty([0,t_f],[-1,1])} \int_0^{t_f} u(t)^2\, dt, \quad t_f > 0 \text{ fixed}$$

$$\begin{cases} \dot{y}(t) = z(t) \\ \dot{z}(t) = u(t) \end{cases} \quad \text{with} \quad \begin{cases} y(0) = 1, \;\; z(0) = 0 \\ y(t_f) = z(t_f) = 0 \end{cases}$$

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Just find $0 \le a < b \le t_f$ OR $0 \le c \le t_f$ numerically!!!

# Existence of solutions to OCP - Examples

Minimal energy reusable rocket landing

$$\min_{u \in L^\infty([0,t_f],U)} \int_0^{t_f} \|u(t)\|^2 \, dt, \quad t_f > 0 \text{ fixed}$$

$$U \triangleq \{u \in \mathbb{R}^3 : \|u\| \leq u_{\max}\} \subseteq \mathbb{R}^3 \qquad \frac{d}{dt}\begin{pmatrix} r \\ v \end{pmatrix}(t) = \begin{pmatrix} v(t) \\ \dfrac{T_{\max} u(t)}{\bar{m}} - g_r \end{pmatrix} \qquad \begin{cases} (r,v)(0) = (r_0, v_0) \\ (r,v)(t_f) = (0,0) \end{cases}$$

<u>It's your turn</u>: By assuming that <u>this OCP has at least one optimal control-trajectory pair</u> $(u^*, x^*)$, apply the PMP and show $u^*(\cdot)$ is of the form (20/30 minutes):

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$$u^*(t) = \begin{cases} \mathfrak{f}(t), & \text{if } \|\mathfrak{f}(t)\| \leq u_{\max} \\ u_{\max} \dfrac{\mathfrak{f}(t)}{\|\mathfrak{f}(t)\|}, & \text{if } \|\mathfrak{f}(t)\| > u_{\max} \end{cases} \qquad \underline{\text{OR}} \qquad u^*(t) = u_{\max} \frac{\mathfrak{f}(t)}{\|\mathfrak{f}(t)\|}$$

<u>Solution</u>: See the blackboard.

# Today’s detailed schedule

1. Necessary conditions for optimality: the Pontryagin Maximum Principle (PMP).

2. Learn how to apply the PMP through simple optimal control problems.

3. Characterize optimal controls for the reusable rocket landing problem.

# Characterize optimal controls for rocket landing

Minimal fuel consumption reusable rocket landing (with fixed or free final time)

$$\min_u \; -m(t_f)$$

$$0 < u_{\min} \leq \|u(t)\| \leq u_{\max}$$

$$\frac{d}{dt}\begin{pmatrix} r \\ v \\ m \end{pmatrix}(t) = \begin{pmatrix} v \\ \frac{T_{\max} u}{m} - g_r \\ -q\|u\| \end{pmatrix}(t) \qquad \begin{aligned} &(r, v, m)(0) = (r_0, v_0, m_0), \\ &(r, v)(t_f) = (0, 0) \end{aligned}$$

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1-2. For the Hamiltonian and the adjoint system we have (see the blackboard):

$$x \triangleq (r, v, m) \in \mathbb{R}^7, \quad u \in \mathbb{R}^3, \quad p \triangleq (p_r, p_v, p_m) \in \mathbb{R}^7, \quad p^0 \in \{0, -1\}$$

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In turn, we have discovered optimal controls are <u>bang-bang</u>:

$$u^*(t) = \begin{cases} u_{\min} \frac{p_v^*(t)}{\|p_v^*(t)\|}, & \Psi(t) < 0 \\ u_{\max} \frac{p_v^*(t)}{\|p_v^*(t)\|}, & \Psi(t) > 0 \end{cases}, \quad \Psi(t) = \frac{\|p_v^*(t)\|}{m(t)} - p_m^*(t)q$$

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      E. Or $\Psi(t)$ <u>is first positive and then negative, and possibly positive again</u>.
   e. We conclude:

$$u^*(t) = \begin{cases} u_{\min}, & \Psi(t) < 0 \\ u_{\max}, & \Psi(t) > 0 \end{cases}$$

$$u^*(t) = \begin{cases} u_{\max} \frac{w_1 + w_2 t}{\|w_1 + w_2 t\|}, & t \in [0, t_1] \\ u_{\min} \frac{w_1 + w_2 t}{\|w_1 + w_2 t\|}, & t \in (t_1, t_2] \\ u_{\max} \frac{w_1 + w_2 t}{\|w_1 + w_2 t\|}, & t \in (t_2, t_f] \end{cases}, \quad \begin{array}{c} 0 \le t_1 \le t_2 \le t_f \\ w_1,\ w_2 \in \mathbb{R}^3 \end{array}$$

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Minimal fuel consumption reusable rocket landing (with fixed or free final time)

$$\min_u \; -m(t_f)$$

$$0 < u_{\min} \leq \|u(t)\| \leq u_{\max}$$

$$\frac{d}{dt}\begin{pmatrix} r \\ v \\ m \end{pmatrix}(t) = \begin{pmatrix} v \\ \frac{u}{m} - g_r \\ -q\|u\| \end{pmatrix}(t) \qquad \begin{aligned} &(r, v, m)(0) = (r_0, v_0, m_0), \\ &(r, v)(t_f) = (0, 0) \end{aligned}$$

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Optimal controls are of the form (under some mild assumptions):

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We have reduced the problem to a finite-dimensional optimization problem: just find $0 \leq t_1 \leq t_2 \leq t_f$ and $w_1,\ w_2 \in \mathbb{R}^3$ numerically!!!

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Next lecture!

# Course schedule

1. Optimal Control Problems (OCP): review of ordinary differential equations; existence of solutions to OCP.

2. Optimality Conditions for OCP: the Maximum Principle and structure of optimal controls; application to reusable rocket landing.

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Bring your laptops with Conda installed!!!

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# End of lecture 2

Questions?
Otherwise, see you in two weeks!